

PriceShield

Service Level Agreement

Data Intelligence for Food Security in The Gambia

Service Levels Agreement

Definitions & Exclusions

"Downtime" means minutes in a calendar month where Azure SQL Database, Power BI, or Azure Data Factory is unreachable or fails to respond, per each service's own SLA. Scheduled Microsoft maintenance doesn't count as downtime. Neither does an outage caused by the client's network, credentials, or a third-party integration outside Microsoft's control.

"Force majeure" (defined as a natural disaster, war, government action) suspends every commitment in this document for its duration.

Agreement Overview

This SLA covers the PriceShield platform: the pipeline, storage, and Power BI dashboard built for Gambia's food-price monitoring, benchmarked against Guinea-Bissau.

Parties: Team 3 (the service provider, contactable at Team3@blackdorf.com) and the Government of the Gambia, represented operationally by Fatou Jallow (the client).

Expected service: a monthly-refreshed dashboard, delivered on one of two contracted architectures — Phase 1 Prototype or Phase 2 Scalable Lakehouse, as defined in Infrastructure Phases below.

Description of Services

Ingestion & orchestration — Azure Data Factory runs one batch pipeline per month (1×1,000 activity run, 1 Data Movement Unit), identical across both phases. Runs beyond that monthly schedule are out of scope.

Event-driven processing — Azure Functions (Flex Consumption, 2,048 MB RAM, up to 10 executions/month) handles exchange-rate API calls, within Azure's free tier, identical across both phases.

Reporting — Power BI Pro, two subscriptions (one admin, one consumption), identical across both phases. Additional viewer seats cost \$14/seat/month and aren't included in the base fee.

Storage and transformation differ by phase: Phase 1 runs on Azure Blob Storage and Azure SQL Database; Phase 2 runs on ADLS Gen2 and Azure Databricks (see Infrastructure Phases).

Scope limits — the monthly cost assumes the capacity parameters of whichever phase is contracted. Exceeding them triggers an overage charge or a change request under CHG-01, not an automatic scope increase.

Exclusions — countries other than The Gambia and Guinea-Bissau, on-demand or ad hoc pipeline runs outside the monthly schedule, and Power BI seats beyond the two included are excluded from the base service in both phases.

Responsibilities

Service provider (Team 3):

- Builds, deploys, and operates whichever phase is currently contracted, within that phase's sized capacity.
- Monitors monthly spend against the active phase's baseline (\$32.57 or \$39.26) and flags the client before any change that would increase cost.
- Responds to and resolves incidents per the severity table (Priority 1–3 / Scope Change) elsewhere in this SLA.
- Owns configuration, security, and access control for all contracted services in both phases.
- Updates the Downtime definition and Platform & Infrastructure SLA section to match the live phase before migrating the client to it.

Client (Government of the Gambia / Fatou Jallow):

- Provides and maintains named user accounts for each licensed Power BI seat.
- Approves and funds the Phase 1 → Phase 2 migration, and any further capacity increase, before it's built, via the change-management process.
- Reports issues through the agreed channel (email to Team3@blackdorf.com) with enough detail to reproduce them.
- Confirms in writing before scope changes such as adding a third country or additional commodities.

Infrastructure Phases

This SLA is delivered on one of two contracted architectures. Phase 1 (Prototype) is the architecture in force at contract signature. Phase 2 (Scalable Lakehouse) is the upgrade path, taken when data volume or transformation load outgrows Phase 1's capacity.

Phase 1 — Prototype, \$32.57/month (\$4.57 Azure infrastructure + \$28.00 Power BI Pro): Azure Data Factory (batch ingestion of external CSV files, ~110 MB), Azure Functions (exchange-rate API calls, within the free tier), Azure Blob Storage (General Purpose v2, Hot

tier, 20 GB, roughly 15 years of headroom), Azure SQL Database (Serverless, Gen5, 1 billed vCore, auto-pauses when idle), Power BI Pro (2 subscriptions).

Phase 2 — Scalable Lakehouse, \$39.26/month (\$11.26 Azure infrastructure + \$28.00 Power BI Pro): Azure Data Factory and Azure Functions unchanged; ADLS Gen2 (Standard, Hot tier, 20 GB + high-priority operations) replaces Blob Storage; Azure Databricks (Jobs Compute, Premium, 1 node, 15 Spark-hours/month) replaces Azure SQL Database as the transformation engine; Power BI Pro unchanged.

Whichever phase is live governs the "Downtime" definition and the Platform & Infrastructure SLA figures elsewhere in this document. Azure SQL Database's 99.99% uptime and credit schedule apply under Phase 1; Azure Databricks' and ADLS Gen2's own Microsoft SLA terms apply once Phase 2 goes live, and Team 3 updates that section with the Phase 2 figures before cutover, not after.

Moving from Phase 1 to Phase 2 is a scope change under CHG-01: it needs a written change request, a migration timeline, and client sign-off before Team 3 builds it. The added \$6.69/month isn't billed automatically.

Severity	Definition	Examples	First Response	Issue Updates	Escalation
1	Dashboard unreachable to all viewers, or the monthly pipeline fails to complete and no refreshed data reaches the dashboard	- Power BI workspace is not updated in time (as per req: ADP-04) - Azure SQL Database does not reflect latest monthly data (as per req: ADP-01) - file,visual exporting not available (.xlsx, csv) (as per REP-03)	Within 4 business hours of detection or notification	Every business day until resolved	Team lead notified within 4 business hours. Fix target: 1 business day
2	A core feature is broken; rest of the dashboard still works	- One country's data missing or outdated (as per req: ADP-02) - commodity outdated (as per req: FEW-01) - cross-country comparison broken (as per req: MON-05)	Within 1 business day	Every 2 business days, or on an agreed interval	Automatic review if unresolved within 3 business days. Assigned to the team member who owns that pipeline stage. Fix target: 3 business days
3	Cosmetic or non-blocking issue; a workaround exists	- Currency Standardisation outdated (as per req: as per RCD-05) - minor filter/slicer bug (as per req: REP-01) - slow PowerBI performance (as per req: REP-04)	Within 2 business days	Weekly, or bundled into the next scheduled release	Logged for the next scheduled release. Fix target: next release

Scope Change	Request falls outside the agreed design is logged	· Add a new country or commodity · schema change · new chart type · extend forecasting to a new series	Acknowledged within 1 working day	Periodic update once scoped and scheduled	Routed to the Change Management process (CHG-01): scoped, timed, written sign-off required before building

Platform & Infrastructure SLAs

Data Storage — Azure SQL Database. **Priority 1**

Commitment: The dataset is stored in Azure SQL Database (General Purpose).

SLA: 99.99% monthly availability, guaranteed by Microsoft (≈52.56 minutes of allowed downtime per calendar year). Microsoft's credit schedule: 10% credit below 99.99% uptime, 25% below 99%, 100% below 95%.

Dashboard Availability — Power BI Service. **Priority 1**

Commitment: The dashboard is available to named viewers holding a Power BI Pro license, or, once the workspace moves to Fabric F64+ capacity, any viewer with a free Power BI account.

SLA: 99.9% monthly availability, guaranteed by Microsoft for Power BI API calls and report rendering. Credit schedule: 10% below 99.9% uptime, 25% below 99%.

Ingestion Pipeline — Data Source, Timing & Processing. **Priority 1**

Commitment: The pipeline fetches new food-price data monthly, processes it, and refreshes the dashboard on a pre-agreed schedule.

Pipeline stages (per the agreed architecture): fetch from source (IMF API / HTTP-linked CSV) → filter to Gambia and Guinea-Bissau only → keep only new entries → standardize units → convert to a common currency (**USD**) → apply inflation adjustment → load to Azure SQL Database → refresh Power BI.

SLA:

- **Source data is expected** by the last calendar day of each month, with a 2-week margin (23:59h on the 14th of the following month) to accommodate late-arriving source files.
- Once the pipeline receives a file, processing (clean → standardize → load) is **expected to be completed within 48 hours**.
- The dashboard reflects the new month's data by the end of those 48 hours at most.
- The pipeline itself runs on Azure Data Factory, which guarantees 99.9% successful pipeline execution and **activity runs starting within 10 minutes of their scheduled trigger**

Core Commitments

1. Automated Ingestion & Country Filtering — SR-01. **Priority 1**

Maps to: REQ-ADP-01, REQ-ADP-03

Commitment: The system ingests every new monthly food-price dataset automatically, without manual processing, and keeps only records for The Gambia and Guinea-Bissau in the production-environment dataset.

Compliance: The system ingests all valid monthly files automatically, within the timing SLA above; the curated dataset contains records from these two countries only.

2. Price Trends & Country Comparison — SR-02. **Priority 1**

Maps to: REQ-MON-01, REQ-MON-02, REQ-MON-03

Commitment: The system presents monthly food-price trends at national level, offers regional drill-down, and compares matching commodities between The Gambia and Guinea-Bissau.

Compliance: Users can analyse every available commodity by country, region, and month; all cross-country comparisons use matching commodities, compatible units, and a common currency; the dashboard meets the 99.9% Power BI availability SLA above.

3. Relative Price-Change Monitoring & Intervention Tracking — SR-03. **Priority 1**

Maps to: REQ-MON-04, REQ-MON-05

Commitment: The system calculates and displays relative percentage price changes and cross-country relative price differences, independent of absolute price level or currency, so users can see whether an intervention (subsidy, import, stock release) actually moved prices.

Compliance: Every valid commodity series shows month-on-month and year-on-year percentage change; every comparable commodity shows the percentage difference between both countries; trend views support before/after comparison around a dated intervention.

4. Missing & Invalid Data Handling — DQR-02. **Priority 2**

Maps to: REQ-RCD-02

Commitment: The system preserves missing prices as explicit nulls and quarantines invalid records instead of deleting them or treating them as zero.

Compliance: The system flags 100% of missing prices explicitly; every quarantined record carries its source-file ID, affected row, failed validation rule, and rejection reason.

5. Minimum Data Requirements for Forecasting — AR-02. **Priority 2**

Maps to: REQ-FEW-02

Commitment: The system generates forecasts only for country–commodity series with at least 24-month continuous historical data and temporal coverage.

Compliance: The system limits forecasts to series with at least 24 valid monthly observations and 70% temporal coverage; excluded series show an explicit reason for ineligibility.

6. Uncertainty for Forecasts & Regression Models — AR-03. Priority 2

Commitment: Every published forecast and regression result reports its own uncertainty and model quality, so Fatou can assess reliability before presenting it to ministers.

Compliance: Every forecast includes a point estimate and prediction interval; every regression result includes its coefficient, confidence interval, p-value, and a model-fit measure such as R^2 .

7. Security & Data Governance — SEC-01. Priority 1

Commitment: Data is encrypted at rest (Azure SQL transparent data encryption) and in transit (TLS). Ministry viewers get read-only dashboard access; only the project team can edit the pipeline or the database. All resources sit in a **single named** Azure region, chosen for data-engineering reasons: Azure's Switzerland North region (Zürich)..

Compliance: Access follows least privilege. Every login is tied to a named individual, not a shared account. Access changes are logged.

8. Change Management — CHG-01. Priority 3

Commitment: A request to add a commodity, a country, or change the schema goes through a written change request. The team scopes the solution, gives a timeline, and gets written sign-off before building. out-of-plan changes aren't built.

9. Exit & Offboarding — EXIT-01 Priority 1

Commitment: If the engagement ends, the team exports the full dataset and the Power BI file to the client within **14** business days, and hands over pipeline source code and documentation sufficient for another team to operate it.

10. Support & Incident Response — SUP-01

Commitment: The project team supports the pipeline and dashboard during business hours (Monday–Friday, 10:00–18:00 CET) and responds by severity.

Severity	Definition	Acknowledge within	Target resolution
Critical (1)	Dashboard unreachable, or the monthly pipeline fails to complete	4 business hours	1 business day
Major (2)	Partial failure — one country's data missing, forecasts unavailable, a data-quality check fails silently	1 business day	3 business days

Minor (3)	Cosmetic or non-blocking issue	2 business days	Next scheduled release
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Compliance: Fatou or delegate reports issues by email to **Team3@blackdorf.com**. The team logs each report with a timestamp and severity, and tracks time-to-acknowledge and time-to-resolution against the targets above.